# A FLAT MAP OF LOCALLY NOETHERIAN SPACES FROM A COHEN-MACAULAY SOURCE IS COHEN-MACAULAY WITH CM TARGET LOCAL RINGS

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*Editorial note.* Statement span is lines 6351-6358 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6360-6365, the complete author proof (4 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* The scheme-level analogue in stacks-more-morphisms (lemma-flat-morphism-from-CM-scheme, lines 5899-5909) has an identical conclusion but its proof is a bare pointer with no reduction written; per the paired-chapter rule only the more general spaces statement is counted.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The translation of the Cohen-Macaulay property for spaces into the corresponding property of Noetherian local rings (lemma-CM, same chapter), the parallel argument in the proof of lemma-composition-CM, and the algebra result Algebra, lemma-CM-goes-up.

*Source reference.* Original lemma 045J: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6365–6365. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 045S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6241–6253. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0C0W: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6311–6326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be a flat morphism of locally Noetherian algebraic spaces over $S$. If $X$ is Cohen-Macaulay, then $f$ is Cohen-Macaulay and $\mathcal{O}_{Y,f(\overline{x})}$ is Cohen-Macaulay for all $x \in |X|$.*

*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Proof.* After translating into algebra using Lemma 045S (compare with the proof of Lemma 0C0W) this follows from Algebra, Lemma 045J. □